

G. Lejeune Dirichlet
Werke, Band II, XXXVIII

Extract from the correspondence between
Alexander von Humboldt and G. Lejeune Dirichlet
English translation; tables and symbols reset in type

It follows from several letters from A. von Humboldt to G. Lejeune Dirichlet that Dirichlet supplied contributions to the *Kosmos*. Thus a letter of Humboldt, which must have been written shortly before the printing of the second volume, that is around 1847, says:

I write these lines to you amid many idle and futile agitations; but it would be a lack of feeling and gratitude on my part not to tell you, my dear and respected friend, how deeply I am touched by the sacrifice of time you made in order to answer my request. You have satisfied me, and perfectly satisfied me, in what alone could enter into the composition of my historical survey. You did it with that measure and precision in the expression of ideas which so highly characterizes everything that comes from your pen. I do not name you; that would make you “the responsible minister”, but I put the passages in quotation marks, which indicates, as I said on p. XIV of the preface, that the sentence is borrowed from one of my friends. Since you are less Indophile than I am, despite duties of kinship, you nurture the malicious hope that one day some fragment of Diophantus will be discovered which will give us what so far is found only among the Indians. I hope this malice will not succeed for you; and if I knew as much mathematics as I do not know, I too would place in the scale of probabilities the particular character of analysis, the local flavor of the Indian algebraists, metrical poets and colorists through symbols. Do not let these phrases make you fear that I have added to the sacramental phrase. I have carefully avoided doing so.

This letter seems to refer to the following passage in the *Kosmos* (vol. II, p. 262):

In the algebraic works of the Indians there is found the general solution of indeterminate equations of the first degree, and a more developed treatment of those of the second degree than in the writings of the Alexandrians that have come down to us; hence there can be no doubt that, if the works of the Indians had become known to Europeans two centuries earlier, and not only in our own day, they would have had a promoting influence on the development of modern analysis.

A. von Humboldt to G. Lejeune Dirichlet

Here, my dear friend and colleague, is my long-announced question concerning the naming of the days of the week by the use of periodic series. The ancients—in the *Somnium Scipionis*, and in a famous passage of Dio Cassius, Ideler vol. I, p. 179—reckoned the planets as follows:

Saturn ♄, Jupiter ♃, Mars ♂, Sun ☉, Venus ♀, Mercury ☿, Moon ☾.

The Jews, however, let the planetary names of the seven-day week follow as Sabbath or Saturday, Sunday, Monday, Tuesday, Wednesday, Thursday, and Friday,

corresponding respectively to Saturn, the Sun, the Moon, Mars, Mercury, Jupiter, and Venus.

There are two widespread solutions of the problem of how the latter sequence arises from the former. According to Dio Cassius' explanation, if one begins to count the hours of day and night from the first hour of the day, dedicating the first to Saturn, the second to Jupiter, and the third to Mars, then the last of the twenty-four hours will be Mars, and the first hour of the day following the Sabbath, Sunday, will fall on the Sun; the third day begins with an hour governed by the Moon. According to Letronne the problem is solved by placing, in the zodiac, twelve signs under thirty-six decans, three decans belonging to each sign. The thirty-six decans are named after the planets in the order of the first table. If one looks at Clerac's *Musee de Sculpture*, p. 832, one finds that the planet which is first in the sign names the weekday:

Leo	♌	♊	♈
Virgo	♍	♎	♏
Libra	♎	♌	♊
Scorpio	♏	♍	♎

The weekdays are therefore ♌, ♍, ♎, ♏, ...

My question is: how can I show briefly and arithmetically that the path of the periodic series 7 and 24 leads to the same result as the same periodic series distributed by threes in 12?

To this question I add another, one that interests me less. The foolish law of planetary distances of Titius—often wrongly called Bode's—is

$$4, \quad 4 + 3, \quad 4 + 6, \quad 4 + 12, \quad 4 + 24, \quad 4 + 48.$$

If Saturn's distance from the Sun is 100, this gives Mercury, Venus, Earth, Mars, the small planets, Jupiter, and Saturn the relative distances

$$\frac{4}{100}, \frac{7}{100}, \frac{10}{100}, \frac{16}{100}, \frac{28}{100}, \frac{52}{100}, \frac{100}{100}.$$

Gauss has already said that it is a mad series which begins with 4, since Mercury should not be 4 but $4 + \frac{1}{4} = 5\frac{1}{4}$. Maedler calls this empirical law a geometrical progression, Bode calls it a harmonic one. It is surely neither. Titius seems to have had dimly in mind the idea of doubling from Mercury onward, not from the Sun onward: 3, 6, 12, 24, ...; and by groping he found for Mercury the constant quantity 4, which, added deformingly to each member of the geometric progression, approximately gives the actual distances. Is that correct? Is not a harmonic progression one in which the first term is to the third as the difference between the first and the second is to the difference between the second and the third,

$$a : b : c : d, \quad a : c = b - a : c - b?$$

Where is anything of that sort to be found in Titius' sequence 4, 7, 10, 16, ..., as Bode wants? $4 : 10 = 3 : 3$. Nonsense.

Do not smile too much at my meagre knowledge. On account of the obscure idea of doubling, I recall the actual distances of the planets from the Sun: Mercury 8 million geographical miles, Venus 15, Earth 20.6, Mars 31.5, small planets 56, Jupiter 107.5, Saturn 197, Uranus 396, Neptune 620.8.

Berlin, Sunday night, 2 o'clock.

With intimate friendship,
A. v. Humboldt.

G. Lejeune Dirichlet to A. von Humboldt

According to two of the three hypotheses proposed for the origin of the planetary weekday names, they arose as follows. In the periodic series formed from the planetary sequence

Saturn *a*, Jupiter *b*, Mars *c*, Sun *d*, Venus *e*, Mercury *f*, Moon *g*,

namely

(1.) *abcdefg, abc...*,

one always took one member and then skipped two, giving the new periodic series

(2.) *adgcfbe, ad...*

This is so whether, according to the first explanation of Dio Cassius, this skipping of two terms arose from the application of the fourth, or whether, as Letronne thinks, one inscribed every three following terms of the first sequence in a sign of the zodiac and used only the first as the weekday name. According to the third hypothesis, Dio Cassius' second explanation, successive weekdays are named after the planets governing the first hour of the day, so that twenty-three terms are always skipped. But in a periodic series such as (1) it is plainly immaterial whether one skips a certain number of terms or that number increased by any multiple of the number of terms in the period, here 7. Hence skipping $23 = 3 \cdot 7 + 2$ must give exactly the same result as skipping 2, namely the series (2).

Concerning Bode's assertion that the Titius series is harmonic, I know nothing else to say than that the term harmonic progression is known to me in no sense other than the one given by Your Excellency, and that nothing else is found in Kluegel either.

Your Excellency's most devoted,
Dirichlet.

Wednesday, 26 May 1851.

The remarks relating to the names of the weekdays were used by Humboldt in the *Kosmos*, where he discusses the question. In the third volume of the *Kosmos*, on pages 474 ff., there is an almost literal printing of the two preceding letters. In the following letter A. von Humboldt informs Dirichlet that he has used his communications.

A. von Humboldt to G. Lejeune Dirichlet

My dear friend! I am in great haste to return a proof to Stuttgart in order to finish this unfortunate third, entirely astronomical, volume. Kindly cast a hurried glance over pages 10 and 11¹ to see whether I have properly grasped your explanation of the agreement of the two methods, that of the planetary hours and that of the 36 decans. If you order changes, give them to me in German and send everything back with the note in your handwriting.

¹The indication of the page number probably refers to the manuscript of the *Kosmos*.